Note: You may use Mathematica or other computer languages to help you solve these problems. If you do, you must show your code and describe your reasoning to get full marks. Do not use generative AI to assist you in this assignment. Please upload your scanned or typeset solutions to Blackboard. The upload should:

- *be a single pdf file. Include any code you have written as part of the PDF.*
- *include your name in the file name as: "surname_student#_assigment#.pdf" and on page 1*

Question 1

Part A (grade: 1-5)

The evolution of the state of a qubit experiencing both coherent driving at frequency Ω and dephasing at rate γ is described by the master equation

$$\frac{d\rho}{dt} = -\frac{i\Omega[\sigma_x, \rho]}{2} + \gamma\, \mathcal{D}[\sigma_z]\rho, \qquad \text{where} \quad \mathcal{D}[a]\rho = a\,\rho a^\dagger - (a^\dagger a\,\rho + \rho a^\dagger a)/2$$

1. Solve for the dynamics of the qubit, assuming an arbitrary initial state.
2. What is the steady state?
3. Does the steady state depend on the initial condition?
4. Assume the initial state is the state $|0\rangle$. Plot the evolution as a function of time (i.e. plot components of the Bloch vector)
5. Plot the evolution of the purity as a function of time,

Part B (Advanced: grade +0,+1,+2)

Use the Master equation for a damped 2-level atom in the interaction picture

$$\frac{d\rho}{dt} = \gamma\frac{(n+1)}{2}\,\mathcal{D}[\sigma^-]\rho + \gamma\frac{n}{2}\,\mathcal{D}[\sigma^+]\rho$$

where $\sigma^+ = |e\rangle\langle g|$ and $\sigma^- = (\sigma^+)^\dagger$.

a) obtain the general solution for ρ for some arbitrary initial state, ρ_0.
b) Does the steady state depend on the initial state?
c) Express the total evolution (from the initial time to the final time) in the Krauss operator-sum representation.

Question 2

The "Steane" code is a 'self-dual' code on 7 physical qubits, in which the X-like stabiliser generators are:

$$S_1 = IIIXXXX$$
$$S_2 = IXXIIXX$$
$$S_3 = XIXIXIX$$

and the Z-like stabilisers, S_4, S_5, S_6 are the same but with the replacement $X \to Z$.

An experimental demonstration of the Steane code in operation was recently shown with an ion-trap quantum computer, see https://arxiv.org/pdf/2107.07505.pdf.

Part A (grade: 1-5)

1. How many logical qubits are encoded in this code?
2. Draw the tanner graph for the stabilisers, and check that the stabilisers commute with each other.
3. Find a pair of operators X_L and Z_L that mutually anti-commute, but commute with all stabilisers. By multiplying the logical operators by stabilisers, the weight of X_L and Z_L can be varied; what is the smallest-weight form of these operators? This is the distance of the code.
4. Assuming an "independent $X-Z$ error model" in which pauli X errors happen with probability p_x, and Z errors happen with probability p_z, describe a "lookup-table" decoding algorithm for the code (you can make reference to the symmetry of this code). Provide a bound on the probability of the decoder failing as a function of p_x and p_z.
5. Can this code correct a single qubit Y error? Check this using the Quantum Error Correction Criteria, and if so, give a practical answer based on your decoder.

Part B (Advanced: grade +0,+1,+2)

a) Calculate the failure rate of the decoder as a function of the bit flip error rate (assuming no phase-flip errors)?

b) Suppose two logical qubits are encoded in two separate instances of the Steane code. Show that the CNOT gate is "transversal"; that is a logical CNOT gate can be effected by application of independent physical CNOT gates between some of the physical qubits. You may find it helpful to use the "Heisenberg picture" transformation laws of stabilisers under a gate.